

PROJECT PRESENTATION



Security

**„SECURITY ANALYSIS OF ARTIFICIAL
INTELLIGENCE MODELS IN THE
CONTEXT OF ADVERSARIAL
ATTACKS AND A COMPARISON OF
IBM ADVERSARIAL ROBUSTNESS
TOOLBOX (ART) AND FOOLBOX”**

Adversarial Attacks



01

➤ minimal changes to input data

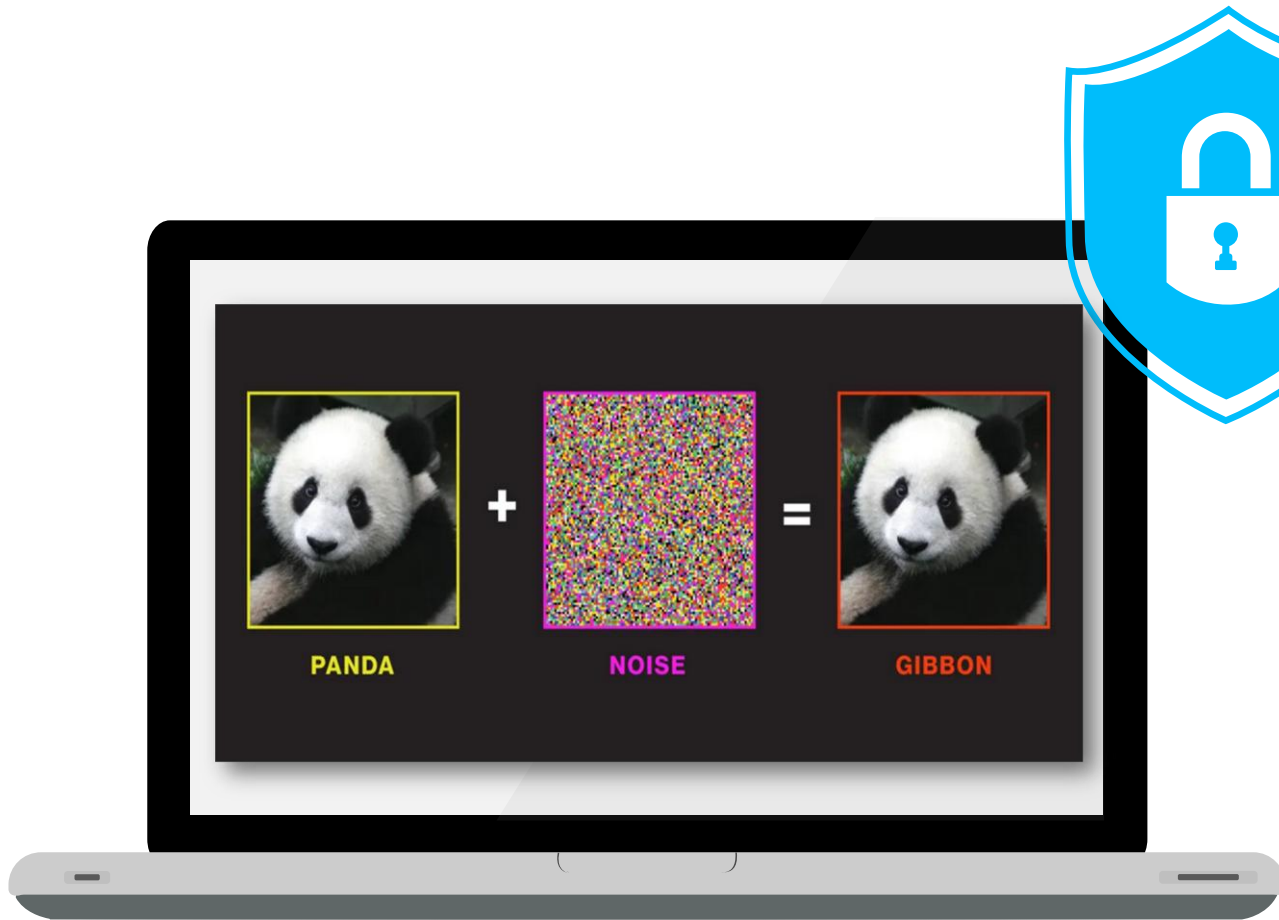
02

➤ lead to incorrect model classification

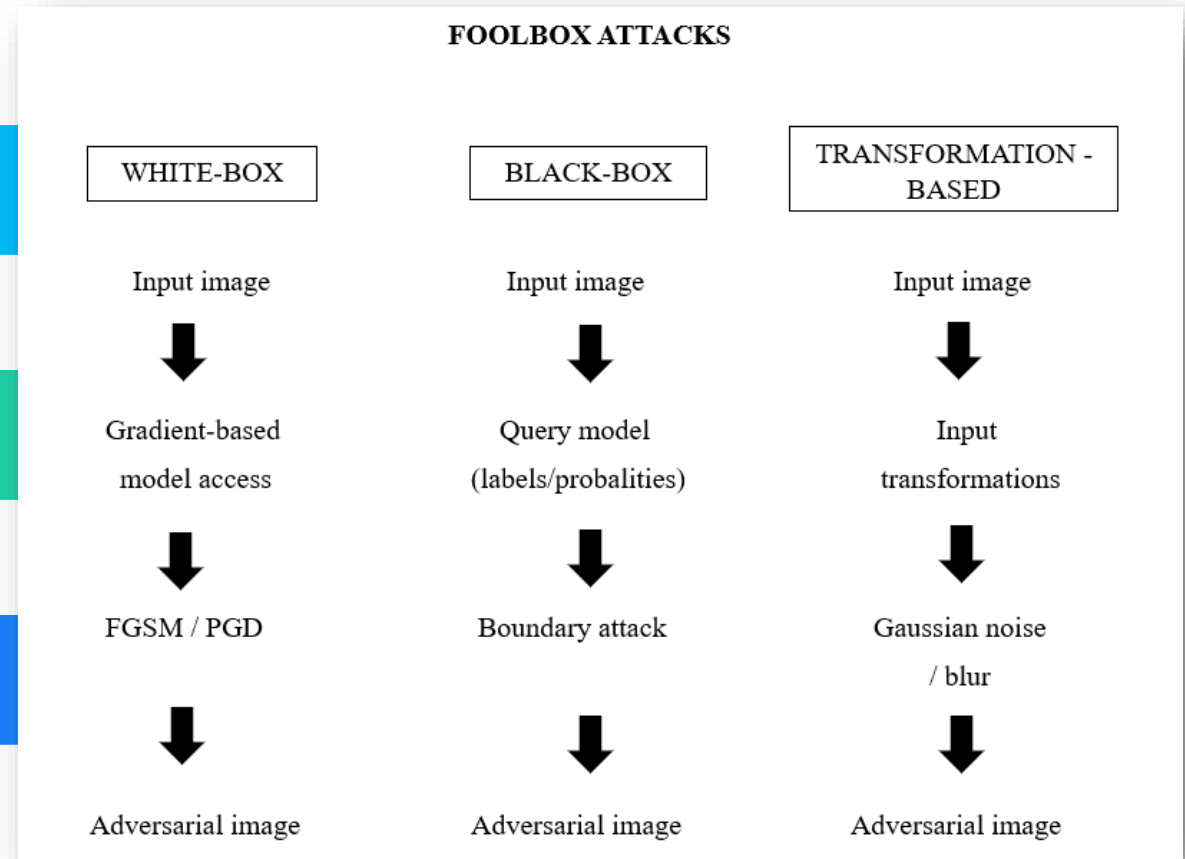
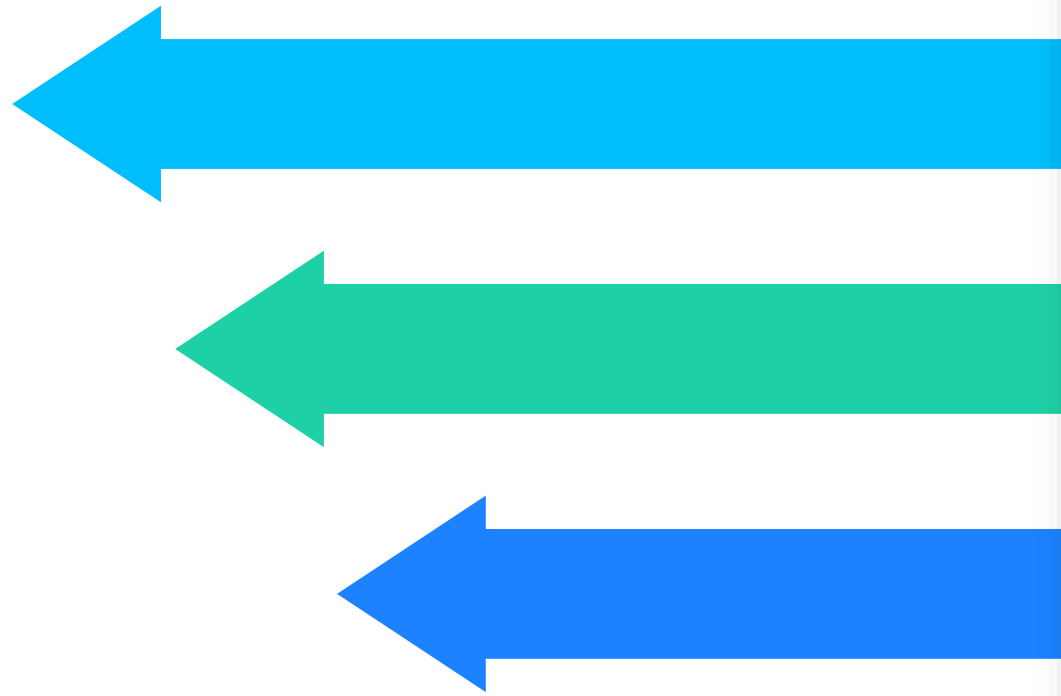
03

➤ difficult to detect

EXAMPLE OF AN ADVERSARIAL ATTACK



FOOLBOX – TYPES OF ADVERSARIAL ATTACKS



MNIST Model Implementation – Test Data Preparation and ART Classifier Configuration

```
# Load MNIST test dataset
dataset = datasets.MNIST(
    root="./data",
    train=False,
    download=True,
    transform=transforms.ToTensor()
)

# Configure ART classifier
classifier = PyTorchClassifier(
    model=model,
    loss=loss_fn,
    optimizer=optimizer,
    input_shape=(1, 28, 28),
    nb_classes=10
)

# Configure FGSM adversarial attack
attack = FastGradientMethod(
    estimator=classifier,
    eps=0.3
)
```

```
eps=0.3
```

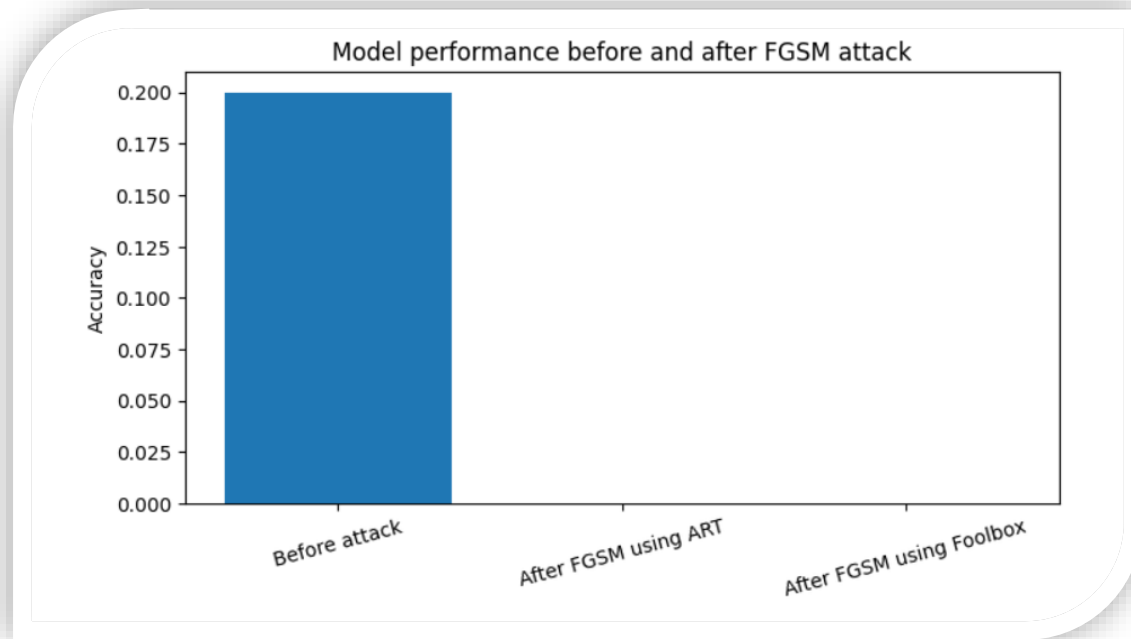
```
estimator=classifier
```

```
attack = FastGradientMethod(
```

```
# Configure FGSM adversarial attack
```



RESULTS – MODEL ACCURACY BEFORE AND AFTER THE FGSM ATTACK



TOOL COMPARISON: FOOLBOX VS ART



Aspect	Foolbox	IBM ART
Difficulty	Low	High
User level	Beginner	Advanced
API	Simple, intuitive	Complex, extensive
Configuration	Simple, partly automated	Manual, many options
Models	CNNs, classifiers	Complex classifiers, ensembles
Frameworks	PyTorch, TensorFlow, Keras	PyTorch, TensorFlow, Keras, Scikit-learn
Features	Minimal perturbation search	Randomized Smoothing, Ensemble Classifier
Usage	Testing, benchmarking	Industrial robustness analysis

REFERENCES

Option E

Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. *Towards Deep Learning Models Resistant to Adversarial Attacks*. (2018).
<https://arxiv.org/abs/1706.06083>

Option D

Foolbox Documentation. *Adversarial Attacks*.
<https://foolbox.readthedocs.io/en/stable/modules/attacks.html>

Option C

gov.pl. *Threats to AI-Based Systems*.
<https://www.gov.pl/web/baza-wiedzy/zagrozenia-dla-systemow-wykorzystujacych-ai>

Option B

becyber.pl. *Adversarial Attacks on AI – What They Are and How to Protect Against Them*.
<https://becyber.pl/ataki-adwersarialne-na-ai-co-to-jest-i-jak-sie-chronic/>

Option A

MIT Technology Review. *How We Might Protect Ourselves from Malicious AI*. (2019).
<https://www.technologyreview.com/2019/05/19/135299/how-we-might-protect-ourselves-from-malicious-ai/>



THANK YOU FOR YOUR ATTENTION